

HAYUN SONG

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CONTACT INFORMATION

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PROFESSIONAL EXPERIENCE

Associate Research Fellow, Korea Institute for International Economic Policy (KIEP) Aug. 2024–Present
International Macroeconomics Team, Department of International Macroeconomics and Finance

EDUCATION

Ph.D., Economics, University of Southern California, CA, US 2024
High-dimensional Bayesian nonparanormal dynamic conditional model with multivariate volatility applications
Committee: M. Hashem Pesaran (Chair), Cheng Hsiao, Timothy Armstrong
M.S., Economics, University of Wisconsin–Madison, WI, US 2016
M.A., Economics, Boston University, MA, US 2014
B.A., Economics, Sungkyunkwan University, Seoul, South Korea 2012
B.A., Business Administration, Sungkyunkwan University, Seoul, South Korea 2012

FIELDS OF INTEREST

Econometrics, Applied Econometrics, Bayesian Statistics

RESEARCH PAPERS

- Song, Hayun (2024). *High-dimensional Bayesian nonparanormal dynamic conditional model with multivariate volatility applications*.
Abstract: This paper develops a Bayesian method for estimating large conditional precision matrices within a DCC–GARCH framework. Wishart likelihoods, horseshoe priors, and a nonparanormal rank transformation allow sparse conditional dependence estimation even when the number of variables exceeds the number of observations. Monte Carlo experiments and applications to international stock indices and asset-pricing models demonstrate the method’s empirical usefulness.
- Lee, Junghyuk and Hayun Song (2023). *Female labor force participation and gender role attitudes*. [Manuscript in preparation].
- Song, Hayun (2021). *Bayesian dynamic factor augmented structure learning: cross-sectional dependence for residuals*.
Abstract: This paper proposes a Bayesian dynamic factor-augmented VAR that represents contemporaneous cross-sectional dependence as a graphical model. Factors are estimated through principal components and forward-filtering backward-sampling, while rescaled spike-and-slab priors address high-dimensional transition matrices. A fractional Bayes factor supports graphical model selection, with simulations and an application to U.S. house prices illustrating the approach.
- Song, Hayun (2018). *Individual heterogeneity in the returns to schooling: instrumental variable quantile regression*.
Abstract: Using instrumental variable quantile regression on twins data, this paper studies heterogeneity in returns to schooling while addressing ability and measurement-error biases. Estimated returns range from 9 to 15 percent across quantiles, and Wald-type location-shift tests provide evidence of individual heterogeneity. The analysis also examines ability bias, measurement error, linearity, and heterogeneous returns to other worker characteristics.

KIEP POLICY RESEARCH AND PUBLICATIONS

- Hayun Song and Munsu Kang (2026). *Oil Price Shocks from the U.S.-Iran War: Spillover Effects on Major Economies*. KIEP Today’s World Economy 26-09.
- Hayun Song and ByeongJun Lee (2026). *AI Usage in Major Economies and Korea*. KIEP World Economy Focus 26-04.
- Hayun Song, Sang-Ha Yoon, and Youngdoo Cho (2026). *Economic Spillovers of Major External Shocks: Trade*

Policy Uncertainty and Oil Prices. KIEP Today's World Economy 26-03.

4. Jiyoung Moon, Hayun Song, Hong Won Kim, JiWon Choi, and Ko Un Cho (2025). *China's Medium- and Long-Term Growth Outlook and Changes in Its Growth Structure*. KIEP Basic Research Report 25-12.
5. Hongseok Choi, Jee Won Park, Hayun Song, ByeongJun Lee, and Kotbee Shin (2025). *Global Trends in High Debt Levels and Their Macroeconomic Implications*. KIEP Basic Research Report 25-11.
6. Jung Eun Yoon, Hayun Song, and Byungjun Lee (2025). *Cross-Country Economic Spillover Effects of Intangible Asset: Technology Diffusion and Implications*. KIEP Policy Analyses 25-01.
7. Song, Hayun (2025). *Small Differences, Large Divisions: How Political Gaps Widen Over Time*. KIEP Opinions 319.

FELLOWSHIPS, HONORS, AND AWARDS

Summer Research Award , Department of Economics, USC	2021
Best third-year paper award.	
Summer Research Award , Department of Economics, USC	2018
Best second-year paper award.	
Scholarship for Excellence , Sungkyunkwan University	2009
Awarded to the top 50 students in the department.	

RESEARCH AND TEACHING ASSISTANCE

Research Assistant , University of Southern California	Fall 2021–2024
Prof. M. Hashem Pesaran, with Prof. Ron Smith	
Teaching Assistant , University of Southern California	
ECON 318: Introduction to Econometrics (undergraduate)	Summer 2022, Spring 2022
ECON 513: Practice of Econometrics (master's)	Fall 2021
ECON 611: Probability and Statistics for Economists (Ph.D.)	Fall 2018
ECON 401: Mathematical Methods in Economics (undergraduate)	Fall 2018
ECON 305: Intermediate Macroeconomic Theory (undergraduate)	Fall 2017, Spring 2018

DATA ANALYTIC PROJECTS

1. **LGBT Clinic Data Analysis** 2018
Analyzed clinic appointment and insurance data, identifying systematic differences in visit timing and insurance coverage across patient groups.
2. **Anti-AgingGames.com Senior Center Cognitive Training Pilot Study** 2018
Assessed the relationship between game participation and word and digit memory scores, and proposed a new game design separating visual and auditory effects.
3. **Psychology Evaluation of Fotonovela**, with Aviroop Ghosal 2018
Evaluated a photo-based educational tool for dementia knowledge and found significant gains in post-intervention and follow-up test scores relative to a standard brochure.

OTHER INFORMATION

Military Service: Corporal, Republic of Korea Army (ROKA) Feb. 2019–2020
Programming Languages and Frameworks: Python (CuPy, PyTorch), R, Stata, Gauss, Git, LaTeX, MATLAB, MySQL, Microsoft Office
Languages: English (Native Fluency), Japanese (Native Fluency), Korean (Native)
Citizenship: South Korea

REFERENCES

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